

Crop Yield Prediction Using Classical Machine Learning Models and Climatic Factors

Adyan Shamim - AU2120040, R. Priscilla - AU2340001, Shreya Dhumal - AU2340124, Vishwa Bharoliya - AU2340002

Team – Vector Minds

School of Engineering and Applied Science, Ahmedabad University, Ahmedabad, Gujarat – 380009

Abstract—Accuracy in Crop yield prediction is very essential in improving agricultural planning, resource allocation, and food security in countries like India. This project presents a machine learning based framework for crop yield prediction which uses historical agricultural data and climate data from Indian states where multiple variables are used in the dataset. Problem formulation was done as a regression task where data preprocessing and feature engineering techniques were applied. Multiple tree-based regression models were implemented including Random Forest, Gradient Boosting, and XGBoost while doing hyperparameter tuning using Bayesian Optimization. In addition, a stacking ensemble model was also developed which combined multiple models to enhance performance. Mean Absolute Error, Root Mean Squared Error, and R^2 score metrics were used for model evaluation and learning curves were analyzed to assess model generalization and residual analysis for evaluating prediction errors. The results clearly demonstrate that the stacking ensemble model had the best overall performance. Furthermore, SHAP was used for key feature identification and interpretation of model predictions. Thus, the developed system provides a robust and interpretable approach for crop yield prediction and has the ability to support data-driven prediction in agriculture.

Index Terms—Machine Learning, Crop Yield Prediction, Random Forest Model, XGBoost, Gradient Boosting, Stacking Ensemble, Bayesian Optimization, Optuna Optimization, SHAP analysis, Feature Engineering, Agriculture Analytics, Climatic Factors, Regression Models, Indian Agriculture, Crop Yield

I. LITERATURE REVIEW

A. Related Work

The main objective of various works on crop yield prediction has been to improve the accuracy using advanced machine learning models, ensemble approaches, and other AI methods.

The study “Yield Prediction for Crops by Gradient-Based Algorithms” demonstrates how effective gradient boosting models are in predicting crop yields based on environmental factors. The results show that these boosting algorithms surpass conventional methods, with CatBoost outperforming other models in terms of accuracy.

Another contribution is “Ensemble Learning Framework for Crop Yield Prediction with Optuna Hyperparameter Tuning”, which combines multiple models to propose an automated ensemble approach. The research employs Optuna for effective hyperparameter tuning and incorporates SHAP-based explainability to point out critical factors resulting in high predictive accuracy and enhanced interpretability.

The study “Coupling Machine Learning and Crop Modeling Improves Crop Yield Prediction in the US Corn Belt” explains a mixed approach that uses machine learning techniques and traditional crop simulation models. This enhances prediction accuracy through capture of both data-driven patterns and biological processes.

Similarly, “Forecasting Corn Yield With Machine Learning Ensembles” is based on using ensemble learning approaches for crop yield prediction. It shows that integrating multiple machine learning models enhances both the prediction accuracy and reliability as compared to individual models, especially in complex datasets.

The research “Random Forests for Global and Regional Crop Yield Predictions” shows the ability of Random Forest models to capture nonlinear relationships between environmental variables and crop yield whilst outperforming conventional regression methods thus, being more suitable for extensive agricultural datasets

B. Contributions

The research works that were reviewed influenced the following design and improvements implemented in our project:

- The use of gradient boosting models such as XGBoost and Gradient Boosting
- The implementation of ensemble techniques (stacking model) based on their improved prediction accuracy and generalization.
- The use of Optuna for hyperparameter tuning as it is observed to show significant enhancement in model performance.
- Incorporation of SHAP explainability aligns with recent work that highlights the importance of interpretability in agricultural machine learning systems.
- The use of Random Forest as a strong baseline model because of its ability to handle nonlinear relationships and diverse agricultural data.
- The emphasis on climatic and agricultural input features aligns with studies showing their critical role in determining crop yield.
- The idea of merging different modeling approaches and improving robustness is shaped by research focused on hybrid and ensemble methods.

II. DATA AND METHODS

A. Dataset

The dataset consists of agricultural data records cultivated across different states in India from year 1997 till 2020. This dataset is taken from kaggle which consists of about 10 columns and 19689 rows in it. The dataset provides crucial information about various features related to crop yield prediction, including crop types, cropping seasons, states, areas under cultivation, annual rainfall, fertilizer used, production quantities, calculated yields and pesticide usage. We have a target variable as crop yield and rest features are factors affecting it in the dataset.

B. Data Preprocessing

While preprocessing the data, duplicate and missing values were removed. The correlations heatmap figure provides key information about fertilizers and pesticides being highly correlated (0.95) and are related to Area (0.97). These are reflecting farm size and provides independent information. For this matter, we applied percentile based capping at the 99th percentile to reduce effect of such extreme values. Then log1p transformation is applied on Area, Fertilizer and Pesticide to reduce skewness. OneHotEncoder is applied to encode categorical variables such as State, Crop and Season. After this, all the numerical features are standardized with StandardScaler. A ColumnTransformer pipeline is used to ensure preprocessing is applied correctly without causing data leakage.

C. Feature Engineering

For better data handling, four additional features were derived from the existing ones. Fertilizer per Area consists of fertilizer efficiency related to a cultivated land. Pesticide per Area again gives information about the pesticide intensity across cultivated land. Rainfall per Area shows the effective rainfall per unit area of cultivation. Input Intensity is the sum of fertilizer and pesticide and captures intensity of agricultural land. These new features improve ability of model to capture relations between climatic and agricultural factors.

D. Models Implemented

There were seven various regression models used for comparison and prediction value. The first one is **Linear Regression** which acts as the basic model. The second model is **Ridge Regression**. This model uses the L2 regularization technique to avoid overfitting problems. **Lasso Regression**, on the other hand, uses L1 regularization and it has capability to simultaneously select the features. **Decision Tree Regressor** is another model that learns the non-linear partition of the training data set by building various trees and calculating the depth of the tree.

E. Hyperparameter Tuning

GridSearchCV with 5-folds cross-validation was performed on Random Forest and Gradient Boosting pipeline with R^2 score as the performance metric. The Bayesian Optimization

algorithm with Optuna library was used to optimize hyperparameters. Random Forest optimization was performed for 40 iterations, and XGBoost for 30 iterations with 5-folds cross-validation.

F. Ensemble and Final Model

A Stacking Ensemble model was built by using Random Forest, XGBoost and Gradient Boosting models as the base models and Ridge Regression model as the meta-model while training the model on 5-fold out-of-fold predictions to avoid any data leakage issue.

G. Evaluation Metrics

The MAE, RMSE, and R^2 Score were used to assess all models against a test set values. The MAE provides average value for prediction error. The RMSE assigns greater penalties for large prediction errors.

III. RESULTS AND DISCUSSION

A. Model Performance Comparison

The Model comparison table given below illustrates that there is a trend common to all the evaluation measures used. The initial Random Forest model provided the best R^2 (log) score of 0.96 accompanied by low MAE values (11) and RMSE (150). This can be considered excellent accuracy considering the untuned nature of the algorithm. Gradient Boosting gave quite low R^2 score of 0.83, demonstrating its high sensitivity to default hyperparameters. Tuned Bayesian models of both algorithms were characterized by similar R^2 and a slight increase in raw MAE and RMSE. As Optuna optimized log-transformed R^2 score, it did not directly minimize the raw error values that can occur despite achieving high accuracy. Finally, the Stacking Ensemble provided R^2 score 0.93 along with MAE 19 and RMSE 265. Since the ensemble of Random Forest, XGBoost, and Gradient Boosting under a meta-model with Ridge regression provided the best overall accuracy, it became our chosen deployment model.

TABLE I
PERFORMANCE COMPARISON OF ALL MODELS

Model	MAE	RMSE	R^2 (log)
Random Forest (Baseline)	11.24	148.32	0.9612
Gradient Boosting (Baseline)	41.38	516.47	0.8301
XGBoost (Baseline)	14.76	204.85	0.9341
RF (Bayesian / Optuna)	15.12	198.63	0.9298
XGB (Bayesian / Optuna)	16.44	251.70	0.9344
Stacking Ensemble	19.21	264.53	0.9312

The baseline Random Forest that the training and validation R^2 converge to high values and balanced performance across all metrics.

B. Feature Importance and SHAP Analysis

Feature importance analysis from the tuned Random Forest shows that Annual Rainfall, Area, Fertilizer per Area, and Pesticide are the most dominant predictors. The engineered per-area features provide additional predictive value, with Annual Rainfall contributing the most.

SHAP analysis further indicates that Fertilizer per Area and Input Intensity are among the most important predictors. These features help capture farming efficiency more effectively than the original input variables.

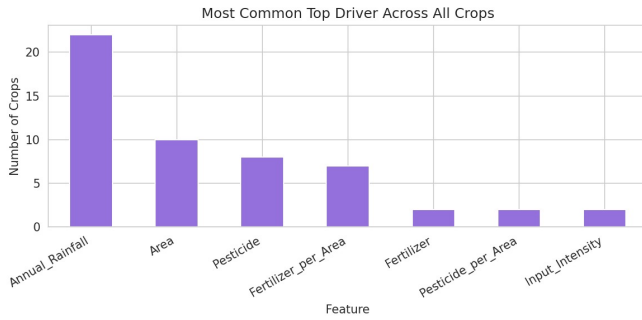


Fig. 1. Feature Importance from Random Forest

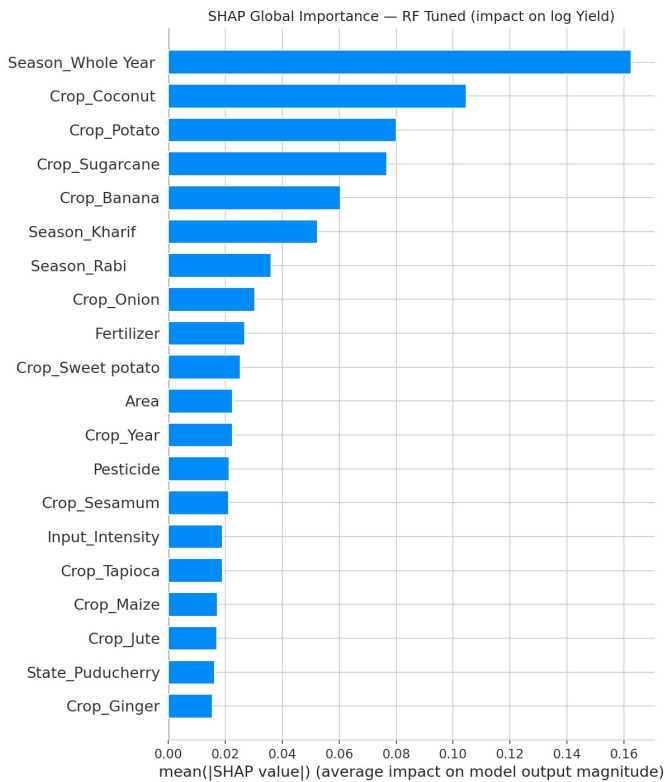


Fig. 2. SHAP Analysis Results

C. Learning Curve Analysis

The learning curve for the tuned Random Forest in figure 4 shows training and validation across R^2 increasing at a high level as training set values increase. This indicates that the model is neither very overfitting nor underfitting. While stacking ensemble figure gives decrease in Training and increasing graph in validation but that upto a certain point. After it crosses 8000 training set value it slightly decreases.

This provides information that the RF tuned is best-fitting model.

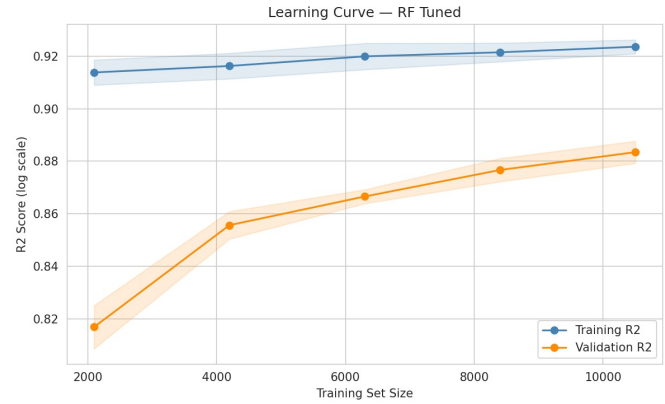


Fig. 3. Learning curve for Random Forest(Tuned)

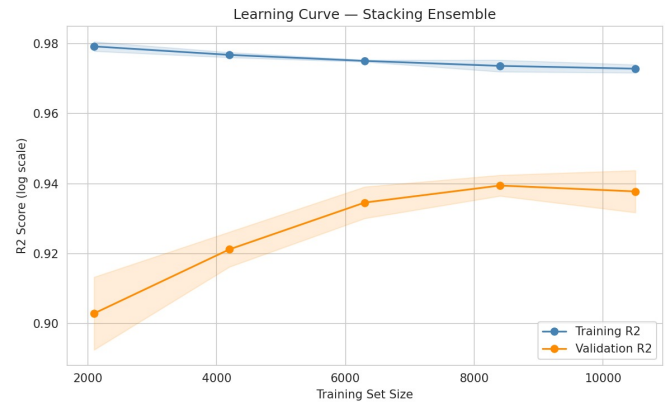


Fig. 4. Learning curve for Staking Ensemble

D. Residual Analysis

The residual plot for the Stacking Ensemble method shown in figure provides information about significant bias in predictions, the residuals are highly concentrated around zero regardless of the predicted value. The distribution of residuals on the logarithmic scale is strongly peaked around zero, implying good model consistency. There is a marginal rise in the variance of residuals at lower values of yield, indicating slightly reduced precision in predictions for low-yield crops. The presence of a few outliers with negative residuals implies some underpredictions for rare cases where crop yields are higher than expected.

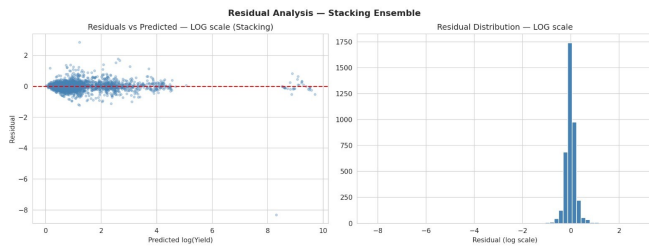


Fig. 5. Actual vs. Predicted crop yield

IV. CONCLUSION AND FUTURE WORK

A. Conclusion

With the help of agricultural and climatic data from India this paper developed and evaluated a complete machine learning pipeline for crop yield prediction. seven classical regression models were implemented which were then compared from linear baselines to ensemble methods. In order to improve model performance a series of methods namely feature engineering, hyperparameter tuning using both Grid-SearchCv and Bayesian Optimization, cross-validation, and stacking ensemble were used. The best performance across all evaluation metrics was displayed by the stacking ensemble method. The Area of cultivation, Annual Rainfall, Fertilizer usage, and Crop type were confirmed to be the dominant predictors through SHAP analyses and feature importance and were consistent with domain knowledge. Additional predictive value was given by the engineered per-area ratio features by capturing farming efficiency. The final model pipeline was saved using joblib.

B. Future Work

additional environmental variables such as soil type, humidity, and temperature can be added into future work. Year-wise yield trends could be captured by applying time-series modelling. The predictive capacity of the pipeline can be improved by extending the pipeline with deep learning approaches and using larger, more geographically diverse datasets. Integration with real-time weather APIs would enable dynamic, season-aware yield forecasting for operational deployment.

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REFERENCES

- [1] J. H. Jeong et al., "Random Forests for Global and Regional Crop Yield Predictions," *PLOS ONE*, vol. 11, no. 6, p. e0156571, Jun. 2016, doi: 10.1371/journal.pone.0156571.
- [2] M. Shahhosseini, G. Hu, and S. V. Archontoulis, "Forecasting Corn Yield With Machine Learning Ensembles," *Frontiers in Plant Science*, vol. 11, Jul. 2020, doi: 10.3389/fpls.2020.01120.
- [3] M. Shahhosseini, G. Hu, I. Huber, and S. V. Archontoulis, "Coupling machine learning and crop modeling improves crop yield prediction in the US Corn Belt," *Scientific Reports*, vol. 11, no. 1, p. 1606, Jan. 2021, doi: 10.1038/s41598-020-80820-1.
- [4] A. Gupta, "Crop Yield in Indian States Dataset," Kaggle, 2023. [Online]. Available: <https://www.kaggle.com/datasets/akshatgupta7/crop-yield-in-indian-states-dataset>.
- [5] S. Otiattakorah, "Agriculture Crop Yield Dataset," Kaggle, 2023. [Online]. Available: <https://www.kaggle.com/datasets/samuelotiattakorah/agriculture-crop-yield>.
- [6] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [7] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama, "Optuna: A Next-generation Hyperparameter Optimization Framework," in *Proc. 25th ACM SIGKDD Int. Conf. Knowledge Discovery & Data Mining*, 2019, pp. 2623–2631.
- [8] P. Mahesh and Rajkumar Soundrapandiyan, "Yield prediction for crops by gradient-based algorithms," *PLoS ONE*, vol. 19, no. 8, pp. e0291928–e0291928, Aug. 2024, doi: <https://doi.org/10.1371/journal.pone.0291928>.
- [9] S. Jayanthi, "Ensemble Learning Framework for Crop Yield Prediction with Optuna Hyperparameter Tuning," *Journal of Information Systems Engineering and Management*, vol. 10, no. 27s, pp. 783–795, Mar. 2025, doi: <https://doi.org/10.52783/jisem.v10i27s.4553>.